A-A-50138
March 31, 1988

COMMERCIAL ITEM DESCRIPTION

FREEZER, MECHANICAL, FOOD

The General Services Administration has authorized the use of this commercial item description in preference to MIL-F-43408.

1. CLASSIFICATION. This commercial item description covers frozen food cabinets of the following types and sizes and classes as specified (see 5.2).

Type	Minimum storage capacity in cubic feet	Approximate dimensions w/o legs (inches)		
		W	D	H
I	20	32	38	85
II	45	59	38	85
III	65	87	38	85

2. SALIENT CHARACTERISTICS.

2.1 Design and construction. The freezers shall be upright models, commercial reach-in types, self-contained and mechanically refrigerated. The freezers and interiors shall be constructed of 20-gauge stainless steel or 20-gauge aluminum with a finish of baked on porcelain enamel. The walls and doors shall be insulated with industrial type fiberglass or expanded polystyrene or preformed or foamed-in-place insulation. Insulation shall provide a "U" factor of 0.0295 BTU/hours/sq.ft./ of temperature difference. Insulation thickness shall not exceed 4 inches. The freezers shall be wired to provide interior lighting when a door is opened. A dial or digital thermometer accurate to $\pm$ 3°F which measures the air temperature in the warmest part of the freezer shall be installed on the top front of each freezer. The freezer shall be equipped with four or six heavy duty adjustable stainless steel legs with a minimum height of 6 inches. Legs shall be provided at or near each corner plus two additional center legs are required for the 65-cubic foot size freezers.

2.2 Refrigeration system. The refrigeration system shall be fully automatic and shall consist of an electrically driven, air cooled condensing unit with evaporator, temperature control, controls for automatic operations, motor starters and protectors, and an automatic defrost. The automatic defrost shall be capable of defrosting when food is stored in cabinet. The system shall be

FSC 4110

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capable of maintaining 10°F or lower in an ambient temperature of 100°F with the compressor running-time not exceeding 75 percent. The system shall also be capable of lowering the temperature of an empty cabinet from 110°F to 0°F in not more than 8 hours. The condensate evaporator shall be capable of removing condensation without the use of a plumbing fixture or floor drain.

2.3 Doors. Type I shall be equipped with two half doors, type II shall be equipped with either four half-doors or two full size doors, and type III shall be equipped with either six half doors or three full size doors. Doors requirement shall be as specified (see 5.2). Replaceable sealing gaskets shall be installed along the inside perimeter of the doors. Anti-sweat heaters shall be installed along the perimeter of each door opening for condensation control.

2.4 Shelves. Type I shall be furnished with three shelves, type II shall be furnished with six shelves, and type III shall be furnished with nine shelves. The shelves shall be wire or sheet metal, stainless steel or steel chrome. They shall be perforated, slotted, molded or stamped. Finish shall be zinc plated, baked on enamel coating or finished aluminum alloy.

2.5 Electrical characteristics. The frozen food cabinets shall be 115 volt or 208/230 volt, 60 hertz, 1-phase a.c. (see 5.2). The controls shall be designed for operation on either a 115 volt or 208/230 volt power source. Unless prohibited (see 5.2) transformers may be used as part of the control system, but shall not be used to step up or step down the voltage for end item use. When a 50 hertz frequency is specified (see 5.2) and the end item is to be operated on that frequency, all systems shall be rated at that frequency. Controls rated at 115 volt, 60 hertz, 1-phase may be used with a 50 Hertz power source, providing the power can be obtained from a hot line and neutral line or through a transformer. Relays, transformers, or any other component necessary for a complete control system shall be furnished as an integral part of the end item. Motors shall conform to NEMA standard MG-1. Cords and outlets shall conform to UL-471.

2.6 Standards compliance. The frozen food cabinets shall comply with Underwriters Laboratories, Inc., Standard No. 471, National Sanitation Foundation Standards No. 7, and National Electrical Manufacturers Association Standard MG-1.

2.7 Identification plate. In addition to the markings required by UL-471, the frozen food cabinets shall be furnished with an identification plate that cites the contract number and national stock number.

3. CERTIFICATION. The contractor shall certify that the product offered meets the salient characteristics and requirements of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same as the product offered for sale in the commercial marketplace. When specific quality assurance provisions are specified for any commercial characteristic the contractor shall furnish data

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resulting from inspection conducted in accordance with the specific quality assurance provisions. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4. PACKAGING.

4.1 Packaging and packing. Packaging and packing shall be commercial or export, as specified (see 5.2).

4.2 Commercial.

4.2.1 Commercial packaging and packing. Each frozen food cabinet shall be cleaned, preserved, packaged, and packed, in accordance with ASTM D 3951.

4.3 Export.

4.3.1 Export packaging. The frozen food cabinet and parts shall be cleaned, preserved, cushioned, and immobilized to protect against corrosion, deterioration, and physical damage during shipment and storage. The legs of the cabinet shall be disassembled and packaged in a domestic fiberboard box constructed and closed in accordance with PPP-B-636. The shelves shall be secured together and packaged in a PPP-B-636 box. Fiberboard boxes containing parts of the cabinet shall be either immobilized within the cabinet or packed into the shipping container along with the cabinet. The outside temperature gauge shall be covered with a fiberboard pad or cushioning material and the pad or material shall be secured in place with pressure-sensitive tape. Doors shall be secured in a closed position with filament reinforced pressure-sensitive tape.

4.3.2 Export packing. Each complete frozen food cabinet shall be packed in a snug-fitting wood-cleated plywood box conforming to domestic type, style A or J, grade B, type 2 load of PPP-B-601 or nailed wooden box conforming to class 1, style 2 or 4, grade B of PPP-B-621, or covered wood crate conforming to type V, style B of MIL-C-52950. The contents of the box or crate, shall be blocked and braced to prevent movement while the container is in transit. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification.

4.4. Labeling (commercial and export). Commercial labeling shall be acceptable.

4.5 Marking (commercial and export). Marking of unit pack and shipping container shall be in accordance with MIL-STD-129.

5. NOTES.

5.1 Intended use. The frozen food cabinets are used in military dining facilities to store food at 0°F or lower.

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5.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type of freezer required (see 2.).
c. Number of doors required (see 2.3).
d. Voltage, phase, operating frequency, when use of a transformer is prohibited, when item is to operate on a 50 hertz power source and power comes from a hot line and a neutral line or through a transformer (see 2.5).

5.3 Source for nongovernment association documents.

UL Standard No. 471 Commercial Refrigeration and Freezers. -- Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062

NSF Standard No. 7 Food Service Refrigerators and Storage Freezers -- National Sanitation Foundation, 3475 Plymouth Road, P.O. Box 1468, Ann Arbor, MI 48106.

NEMA Standard MG-1, National Electrical Manufacturers Association
2101 L Street, N.W.
Washington, D.C. 20037

ASTM D 3951 Standard Practice for Commercial Packaging - American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103

MILITARY CUSTODIANS:

Custodians

Army - GL
Navy - YD
Air Force - 99

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